

Basic Language of Number Theory

1 Big Picture

What number theory gives us

Number theory studies integers. For algorithms, the core language is:

divisibility $\longrightarrow$ primes $\longrightarrow$ remainders $\longrightarrow$ modular classes.

Symbol	Meaning
$\mathbb{Z}$	$\{\dots, -2, -1, 0, 1, 2, \dots\}$, all integers.
$\mathbb{N}$	$\{0, 1, 2, \dots\}$, natural numbers.
d	Usually a divisor.
n	Usually a positive modulus.
q, r	Quotient and remainder.

2 Divisibility

Exact Multiple

$$d \mid a \iff a = kd \text{ for some } k \in \mathbb{Z}.$$

$$3 \mid 12$$

$$12 = 4 \cdot 3$$

no remainder

Not divide

$d \nmid a$ means d does not divide a .

Example: $5 \nmid 12$

Divides zero

Every integer divides 0.

Reason: $0 = 0 \cdot d$

Sign does not matter

$$d \mid a \iff -d \mid a$$

Example: $3 \mid 12, -3 \mid 12$

Divisor bound

If $a \neq 0$, then $1 \leq d \leq |a|$.

Example: each divisor of 24 is ≤ 24 .

Useful rules

If $d \mid a$, $d \mid b$, and $m \in \mathbb{Z}$, then

$$d \mid (a + b), \quad d \mid (a - b), \quad d \mid ma.$$

3 Divisors and Factors

Positive divisors

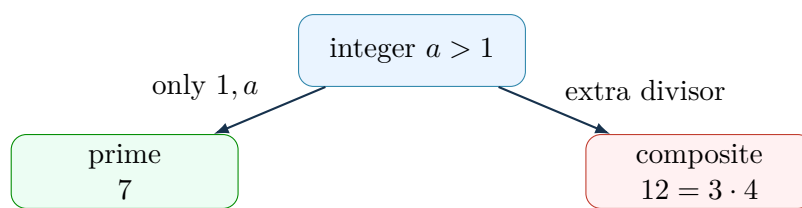
If $d \mid a$ and $d > 0$, then d is a positive divisor of a .

Integer	Positive divisors / factors
24	1, 2, 3, 4, 6, 8, 12, 24.
20	Trivial divisors: 1, 20. Nontrivial factors: 2, 4, 5, 10.

4 Prime and Composite

Classification for $a > 1$

a prime $\iff$ only positive divisors are 1, a ,
 a composite $\iff$ a has an extra positive divisor.



Type	Examples
Prime	2, 3, 5, 7, 11, 13, 17, 19, 23, 29.
Composite	$4 = 2 \cdot 2$, $6 = 2 \cdot 3$, $9 = 3 \cdot 3$, $39 = 3 \cdot 13$.
Neither	1, 0, and negative integers.

Remember

1 is not prime. 2 is the only even prime.

5 Division Theorem

Quotient and Remainder

For $a \in \mathbb{Z}$ and $n > 0$,

$$a = qn + r, \quad 0 \leq r < n.$$

Here q is the quotient and $r = a \bmod n$ is the remainder.

Divisibility test

$$n \mid a \iff a \bmod n = 0.$$

Quick check

so

$$37 = 7 \cdot 5 + 2$$

$$37 \bmod 5 = 2.$$

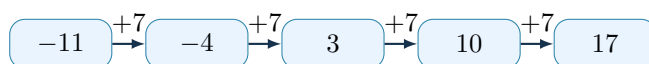
6 Modular Equivalence

Same Remainder

$$a \equiv b \pmod{n} \iff n \mid (a - b).$$

Class modulo n

$$[a]_n = \{a + kn : k \in \mathbb{Z}\}.$$



$$[3]_7 = \{\dots, -11, -4, 3, 10, 17, \dots\}$$

7 $\mathbb{Z}_n$

Residue Classes

$$\mathbb{Z}_n = \{[0]_n, [1]_n, \dots, [n-1]_n\}.$$

Short notation:

$$\mathbb{Z}_n = \{0, 1, \dots, n-1\}.$$

Each number represents a whole class. For example,

$$-1 \equiv n-1 \pmod{n},$$

so -1 represents $[n-1]_n$.

8 Final Summary

Keep These

$$d \mid a \iff a = kd, \quad k \in \mathbb{Z}.$$

$$p \text{ prime} \iff p > 1 \text{ and divisors are only } 1, p.$$

$$a = qn + r, \quad 0 \leq r < n, \quad r = a \bmod n.$$

$$a \equiv b \pmod{n} \iff n \mid (a - b).$$

$$\mathbb{Z}_n = \{[0]_n, [1]_n, \dots, [n-1]_n\}.$$